Revision Checklist for A* vs CoT Paper

Core revision principle: Reframe the paper from "A* is better than CoT" to "A* and CoT are complementary, and search helps when the problem topology rewards branching."

0. Global framing change - do this first

Current weakness

The submitted paper still reads partly like: "we propose A* search and prove admissibility." Reviewers pushed back because the router is weak, the heuristic seems task-dependent, compute is high, and the experiments are limited.

Revised framing

Make the paper about:

When does search help LLM reasoning? CoT and A solve complementary instances; this complementarity is explained by reasoning topology; topology-aware routing is more promising than post-hoc trace judging.*

Where to revise

Title, Abstract, Introduction, Contributions, Discussion, and Conclusion.

Exact action

Change the title to something closer to one of the following:

- When Does Search Help LLM Reasoning? A Topology-Driven Analysis of A* Search versus Chain-of-Thought
- Beyond Linear Reasoning: A Topology-Driven Study of Chain-of-Thought and A* Search

1. Abstract

What to add

- CoT/A* complementarity.
- Reasoning Topology Framework with 13 pre-generation structural features.
- Topology clusters showing A* advantage increases with branching complexity.
- Topology router outperforming output-based routers.
- Fluency-Correctness Asymmetry explaining why LLM-as-critic routing is limited.
- New generalization, sensitivity, and compute experiments.

Where

Replace the current abstract lines that mainly emphasize admissibility and routing with a topology-centered abstract. Keep the point that routing improves but recovers only part of the oracle gain; now explain why.

Suggested abstract skeleton

Chain-of-thought prompting commits an LLM to a single linear reasoning path, while search-based methods explore multiple candidate traces. We study whether guided A search should replace or complement CoT. Across StrategyQA, LogicQA, HotpotQA, and additional math benchmarks, we find that neither method uniformly dominates; instead, they solve partially different subsets of instances. We introduce a Reasoning Topology Framework using 13 pre-generation structural features and show that A* gains increase with branching complexity rather than dataset identity. A topology-aware router that observes no generated traces outperforms output-based routers such as semantic entropy and LLM-as-critic adjudication, recovering a larger fraction of the oracle gap. We further show that post-hoc routing is limited by a Fluency-Correctness Asymmetry: in many disagreement cases, the more fluent trace is wrong. Sensitivity and token-cost analyses clarify when the additional search budget is worthwhile. Overall, A* is not a universal replacement for CoT; it is a higher-cost, topology-sensitive reasoning strategy.*

2. Introduction

2.1 Add a new motivation paragraph after the CoT limitation paragraph

Current location

After the paragraph explaining that CoT is linear and can propagate early mistakes.

What to add

Search-based reasoning relaxes the linearity assumption, but exploration is not free. Search can preserve useful alternatives, but it can also introduce over-exploration, redundant decomposition, spurious branches, and unnecessary reasoning on problems that are already solvable by a compact linear chain. Therefore, the important question is not simply whether search is stronger than CoT, but when the input contains useful branching structure.

2.2 Replace the old research question

Current wording

The paper currently says the focus is whether search and CoT are complementary and whether simple routers exploit that complementarity.

Replace with four explicit questions

- (i) Does guided search dominate linear CoT under a controlled comparison?
- (ii) If not, do the two methods succeed on complementary instances?
- (iii) Can this complementarity be explained by pre-generation structural properties of the input?
- (iv) Can those structural properties be used to route between CoT and A* before any reasoning trace is generated?

2.3 Add a "Why A*?" paragraph

Reviewer concern addressed

Reviewers asked for clearer comparison with ToT and other search-based reasoning methods.

Where

End of Introduction, before Contributions.

Suggested wording

We use A not because it is the universally best search algorithm for LLM reasoning, but because it exposes the minimal ingredients of guided search: a frontier of partial reasoning states, an accumulated path cost, an estimated remaining cost, a branching budget, and a stopping rule. DFS and BFS isolate depth or breadth but lack heuristic prioritization; Tree-of-Thoughts and beam-style methods introduce evaluator-based pruning but entangle decomposition prompts, local scoring, and breadth choices; MCTS-style methods add rollout and value-estimation design choices. A* gives a compact diagnostic abstraction for asking when guided exploration is useful.*

2.4 Rewrite the Contributions

Current problem

The current contribution says the proposed A* is "theoretically superior" because of admissibility. That is risky because the revised analysis uses a critic-based heuristic, for which classical admissibility is not guaranteed in the same way.

Replace with

- Controlled comparison: compare CoT and guided A* across structurally diverse reasoning benchmarks and show that neither uniformly dominates.
- Complementarity: show that CoT and A* solve non-trivially different subsets of instances, making routing a real problem rather than a post-hoc add-on.
- Reasoning topology: introduce 13 pre-generation structural features and show that branching complexity predicts A* advantage better than dataset identity.

- Routing and diagnostics: show that topology-aware routing outperforms output-based routing, and analyze why trace-level adjudication fails through Fluency-Correctness Asymmetry, overthinking failures, sensitivity, and compute-cost analyses.

3. Related Work

3.1 Expand Related Work into five focused paragraphs

Replace the current short related work section with the following structure:

Paragraph 1: CoT and structured reasoning

Include Chain-of-Thought, Self-Consistency, Tree-of-Thoughts, Graph-of-Thoughts, CoT-decoding, and recent work showing that CoT is not universally beneficial. Position your contribution as the search-side complement: identifying when search helps rather than assuming it always helps.

Paragraph 2: Search-augmented reasoning

Include MCTS-based reasoning, LATS, MCTSr, self-evaluation guided beam search, Coconut, and rStar-Math. Emphasize that your work diagnoses when search helps rather than only reporting aggregate gains.

Paragraph 3: Over-exploration and search failures

Include Fetch / redundant-state over-exploration, the overthinking survey, and CoT-Valve. State that prior work often treats over-exploration as an efficiency issue; your paper treats it as both an efficiency and correctness issue.

Paragraph 4: Routing and adaptive computation

Discuss semantic entropy, verifier-based routing, and the distinction from mixture-of-experts routing. Clarify that your router selects between inference-time algorithms, not neural modules.

Paragraph 5: LLM-as-judge and critic limitations

Discuss LLM-as-judge, process supervision, and your Fluency-Correctness Asymmetry. Explain why trace-level judging is insufficient: the more fluent trace can be wrong.

3.2 Add a methodological comparison table

Place this at the end of Related Work or the beginning of Method.

Method	Search structure	Uses frontier?	Uses heuristic?	Pre-generation routing?	Main limitation
CoT	linear	no	no	no	premature commitment
Self-Consistency	multiple linear traces	no	no	no	unguided sampling
ToT	tree/BFS-style	yes	evaluator/pruning	no	breadth/evaluator entangled
MCTS/LATS	tree + rollout	yes	value/rollout	no	many design choices
A*	priority frontier	yes	yes	yes, in this paper	higher compute, topology-sensitive

4. Methodology

4.1 Fix the heuristic/admissibility issue carefully

Critical issue

Do not mix the old depth/coverage theorem with the new critic-based heuristic as if the theorem proves the full system. The original paper proves admissibility for a depth/coverage heuristic under explicit assumptions. The revised version should make clear that the main experimental heuristic is critic-based.

Where

Section 2.4 / Heuristic Design.

Revision plan

Make the critic-based heuristic the main experimental heuristic: h(n) = 1 - score(n), where the critic scores factual consistency and reasoning completeness.

Then add a subsection titled "Approximate Conservative Guidance" with three sufficient cases:

- Monotone calibrated scoring.
- Conservative estimation.
- Bounded overestimation.

Report the empirical validation:

- 500 annotated intermediate reasoning states.
- 100 StrategyQA, 100 HotpotQA, and 300 LogicQA.
- Critic mean score: 0.43.
- Human-assessed progress: 0.58.
- Paired t-test $p < 0.001$.
- Overestimation rate: 12.3%.
- Maximum overestimation: epsilon_max = 0.14.

Move old theorem

Move the original depth/coverage admissibility theorem to an appendix titled "Appendix B: Auxiliary Conservative Heuristic Construction." Do not present it as proving the full critic-based A* system.

4.2 Add "Reasoning Topology Features" subsection

Place this after the A* method and before routing, or after the experimental setup.

Feature	Symbol	Description
Question length	q_len	token count of question
Context length	c_len	token count of full context
Context sentences	c_sent	number of sentences
Logical connectives	conn	count of if, then, therefore, because, unless
Negation markers	neg	count of not, no, never, neither
Comparatives	cmp	count of more, less, better, worse, than
Question marks	q?	number of question marks
Estimated hop count	hop	dependency/parser-based proxy
Contextual density	con	entities per sentence
Distractors	dest	number of candidate options
Branching coefficient	bc	c_sent x hop / q_len
Reasoning depth	ell	maximum dependency depth
Number of options	nopt	answer options available

Add the following sentence: These features are computed before any model generation and are used only to characterize the structural demands of the input.

4.3 Revise routing subsection

Split routers into two families:

Output-based routers

- Semantic Entropy
- LLM-as-Judge
- LLM-as-Critic

Pre-generation routers

- Random Forest over simple features
- Topology Router over 13 topology features

Add this sentence: The topology router should be read not as a final deployed router, but as a diagnostic test of whether the CoT/A boundary is predictable from problem structure before reasoning begins.*

5. Experimental Setup

5.1 Harmonize dataset sizes before adding anything

Critical warning: the original ARR paper and the TMLR draft appear to use different dataset sizes/protocols. Before revision, choose one protocol and make every table consistent.

[ ] Same dataset sizes in Abstract, Setup, Table 1, complementarity table, topology clustering, topic modelling, router table, and appendix.

[ ] Same model for CoT and A* in each comparison.

[ ] Same split used for topology clustering and router cross-validation.

[ ] If topology/router results are only on a subset, state this clearly.

[ ] Do not paste TMLR numbers into the ARR paper without aligning splits.

5.2 Add larger-model experiment

Place this in the Results section, after the main table. Use "14B-scale" rather than "13B" if the actual model is qwen2.5:14b.

Dataset	CoT	A*	Interpretation
StrategyQA	76.02	75.56	search not uniformly better
HotpotQA	60.19	83.33	search strongly helps multi-hop evidence chaining
LogicQA	57.91	58.12	roughly tied

Suggested text: These results show that the pattern is not restricted to 0.5B/8B models. The sign of the A gain still depends on the reasoning structure: large gains appear on multi-hop evidence chaining, while compact or direct settings do not uniformly benefit from search.*

5.3 Add math-domain extension

Place this after the 14B model table.

Dataset	CoT	A*
GSM8K	85.04	86.12
MATH	78.00	79.10

Suggested text: A gives modest gains on GSM8K and MATH, but the gains are smaller than in topology-heavy multi-hop QA. This supports the central claim that search is most useful when the problem benefits from maintaining alternatives, not merely because the benchmark is difficult. Coding and long-form reasoning should be stated as future work, not solved.*

6. Main Results

6.1 Update Table 1

Separate the current table into two clearer tables.

Table 1: Base reasoning methods

Columns: Dataset, CoT, A*, Self-Consistency, ToT, Oracle, Oracle gap over best single method.

Table 2: Routers

Columns: Router, input used, StrategyQA, LogicQA, HotpotQA if available, and gap recovery.

6.2 Add "From aggregate inconsistency to complementarity"

Place this immediately after Table 1.

Suggested text: If one method dominated and the oracle added little, there would be no meaningful routing problem. The presence of substantial CoT-only and A-only cells shows that the methods implement different useful computations.*

Add the four-cell complementarity table using the final chosen protocol:

Dataset	Both correct	CoT only	A* only	Both wrong
StrategyQA	...	...	...	...
LogicQA	...	...	...	...
HotpotQA	...	...	...	...

7. New Section: Explaining Complementarity Through Topology

This should be the central new section.

7.1 Add "Reasoning Topology: Four Cluster Types"

Place this after the complementarity table and before routing.

Cluster	Profile	BC	A*	CoT	Both%	Interpretation
C1	Linear, short	1.28	81.7	82.1	76.4	CoT sufficient
C2	Mid-branch	4.24	73.5	66.2	57.8	selection critical
C3	Complex, noisy	4.02	52.3	49.1	38.6	both often fail
C4	Deep-branch	6.49	69.8	58.4	47.2	A* advantage largest

Add a figure: A* advantage versus branching coefficient, with linear fit R^2 = 0.94.

Suggested text: As branching complexity increases from C1 to C4, A advantage increases from approximately -0.4 pp to +11.4 pp. The clusters are not dataset labels: each contains instances from multiple benchmarks. This indicates that the CoT/A* boundary cuts across benchmark identity and is better explained by problem structure.*

7.2 Add "Prescriptive decision rule"

Place this at the end of the topology cluster subsection.

- C1 / linear: use CoT; search overhead is unnecessary.
- C2 / mid-branch: route adaptively; this is where router quality matters most.
- C3 / complex noisy: neither method is reliable; retrieval or stronger models may be needed.
- C4 / deep-branch: use A*; search has the clearest advantage.

7.3 Add "Topic modelling validates topology"

Place this after cluster analysis.

Category group	Topics	Coverage
A*-dominant	content-domain factual; assumption analysis; temporal event; argument-effectiveness evaluation; biography factual	81.4%
CoT-dominant	comparison factual; statement-strength evaluation; best-option selection	18.6%

Suggested text: A-dominant topics share multi-hop chaining, premise scanning, temporal ordering, or entity resolution. CoT-dominant topics are mostly single-step inference categories.*

8. New Section: Topology-Aware Routing

8.1 Replace "router is weak" framing

Reviewer concern addressed: routing is weak and the strongest router is hand-crafted. Your answer should be: yes, and this is the point - trace-based routing is limited, topology-based pre-routing is stronger.

Router	Input	StrategyQA	LogicQA	Gap recovered
A* only	-	74.80	44.85	-
CoT only	-	64.52	45.78	-
Semantic Entropy	traces/samples	74.37	47.47	14.8%
LLM-as-Critic	traces	74.19	47.16	12.4%
Random Forest	simple features	76.25	48.20	18.2%
Topology Router	13 features	77.35	49.01	26.1%
Oracle	-	84.59	59.29	100%

Suggested text: The topology router observes no generated traces, yet outperforms output-based routers. This reverses the intuitive expectation that seeing full traces should be more informative. The bottleneck is often not judging which completed trace looks better, but recognizing before generation which computation the problem requires.

8.2 Add Fluency-Correctness Asymmetry

Place this immediately after the router table.

Metric	A*	CoT
Average fluency score	0.725	0.673
Cases where more fluent	62.0%	38.0%
More-fluent but wrong	38.0% overall	38.0% overall

Suggested text: Output-based routers are limited because trace fluency is not equivalent to trace correctness. In a substantial fraction of disagreement cases, CoT produces the more fluent trace but the wrong answer. A critic that rewards coherence can therefore prefer the wrong computation.

9. Sensitivity Analysis

9.1 Add budget sweep

Place in the main paper if space permits; otherwise put in Appendix D with a 2-3 sentence summary in Results.

Budget B	5	10	15	20	30
Accuracy	73.0	77.3	78.0	78.7	76.0

Suggested text: Accuracy improves from B=5 to B=20, then drops at B=30, suggesting diminishing returns and over-exploration. The default B=15 lies within 0.7 pp of the empirical peak while using less compute than B=20.

9.2 Add branching factor sweep

Branching K	1	2	3	4
Accuracy	77.7	78.3	76.0	80.3

Suggested text: The K=1 variant degrades gracefully and already exceeds self-consistency on the same sample, while larger branching can help but is not monotonic. This supports the claim that search budget must be matched to topology.

10. Compute-Accuracy Tradeoff

10.1 Add a dedicated compute table

Place a compact version in the main paper because reviewers explicitly asked for compute-performance tradeoff.

Method	Accuracy	Tokens/question	Notes
CoT	76.80	~512	single call
SC-5	78.21	~2,560	5 CoT traces
A*, K=1, B=30	77.7	3,372	linear/no branching
A*, K=3, B=15	78.0	12,452	default
A*, K=3, B=20	78.7	12,391	peak sample accuracy

Important wording: A is substantially more expensive than CoT and SC. Its use is justified only when topology predicts that branching is useful. This motivates topology-aware pre-routing as a compute-saving mechanism.*

11. Failure Analysis

11.1 Add A* overthinking failure taxonomy

Place in Discussion or Appendix, but include at least the summary table in the main paper.

Failure mode	Proportion	Description
Overthinking	91.1%	search continues past correct evidence and accumulates distractors
Premature commitment	8.9%	search locks onto an early branch

Add 2-3 representative examples from the TMLR draft, such as Disney ice princess, Menthol/Thanksgiving, and Albany GA vs Albany NY.

Suggested text: CoT often fails by committing too early; A often fails by continuing too long. Thus, the same branching mechanism that helps A* on multi-hop and premise-scanning problems hurts it on linear problems.*

12. DeepSeek / LLM judge dependence

12.1 Add explanation in Method

Place this in the Method subsection on LLM-as-Critic.

We use DeepSeek-R1-Distill-Qwen-14B because it is a strong open instruction-following model for reasoning-trace critique, is architecturally distinct from the LLaMA-3.1-8B generation backbone, and can be run locally without closed API dependence. However, we do not make DeepSeek the core contribution: in the revised analysis, the LLM critic is treated as an output-based routing baseline, while the main routing result is the topology router, which uses no LLM inference.

If time permits, add one independent critic-model ablation. If not, explicitly state: Judge sensitivity remains a limitation, but the topology router reduces reliance on LLM-as-judge routing.

13. Broader benchmarks and domains

13.1 Add GSM8K and MATH, but with restraint

Place this in a Results subsection titled "Extension beyond QA."

Suggested stance: These results broaden the evaluation beyond QA, but they do not settle coding or long-form reasoning. We therefore treat coding and long-form reasoning as future work.

14. Discussion

14.1 Rewrite Discussion around three takeaways

Takeaway 1: Search is topology-sensitive

Search helps when the problem requires preserving alternatives, scanning premises, chaining facts, or resolving entity paths. CoT is preferable when the answer follows from a compact linear inference.

Takeaway 2: Routing should happen before generation

Post-hoc trace inspection is limited because fluent traces can be wrong. The topology router success suggests that adaptive inference should begin with problem-structure recognition.

Takeaway 3: Topology is necessary but not sufficient

The topology router recovers 26.1% of the oracle gap, leaving substantial residual error. Closing this gap likely requires modeling problem-model compatibility: whether the base model has the parametric knowledge and representational capacity needed to exploit the search space.

15. Limitations

Replace the current limitations with sharper, more honest ones:

- Search-policy dependence: results are based on one budgeted A* implementation; other search policies may shift the boundary.
- Hand-crafted topology features: the 13 features are proxies, not latent causal topology.
- Single-language setting: dependency/parser-based features may not transfer directly to non-English settings.
- Judge/critic dependence: critic-based heuristic uses DeepSeek; topology router reduces but does not eliminate judge-related concerns.
- Compute cost: A* is more expensive; token accounting is measured on ablation samples and should be expanded.
- Domain coverage: math is added, but coding and long-form reasoning remain future work.
- No learned router: future work should learn topology embeddings or jointly train routing/search policies.

16. Appendix organization

Appendix A: Detailed qualitative examples

Move existing examples here, plus the new overthinking examples.

Appendix B: Auxiliary conservative heuristic proof

Move the old admissibility theorem here. Clearly say it applies to the depth-coverage heuristic, not the critic-based heuristic.

Appendix C: Prompts

Include CoT prompt, A* step-generation prompt, DeepSeek critic prompt, LLM-as-judge prompt, and topology feature extraction details.

Appendix D: Sensitivity analysis

Budget and branching sweeps.

Appendix E: Compute analysis

Token counts and node counts.

Appendix F: Additional models/domains

14B-scale results, GSM8K, and MATH.

Appendix G: Reproducibility details

Exact dataset splits, seeds, model versions, Ollama/vLLM settings, hardware, and code path.

17. Must-not-miss consistency checks

[ ] Do not claim the critic-based heuristic has the same formal admissibility guarantee as the depth-coverage heuristic.

[ ] Do not call A* "theoretically superior" to standard search; say it is a controlled diagnostic abstraction.

[ ] Do not mix dataset sizes between ARR and TMLR versions.

[ ] Do not call the topology router fully scalable or learned; it is still hand-crafted, but more principled and pre-generation.

[ ] Do not say the router closes the oracle gap; say it recovers only part of it.

[ ] Do not overclaim math/coding generalization; math is modestly tested, coding/long-form remain future work.

[ ] Label all subset experiments clearly: full dataset vs. 300-question HotpotQA sample vs. matched 14B subset.

[ ] Add confidence intervals/statistical tests wherever possible, especially for new tables.

[ ] Update abstract, conclusion, and limitations so they all tell the same topology-sensitive story.

Highest-priority revision order

1. Reframe Introduction + Contributions around topology.
2. Fix heuristic/admissibility presentation.
3. Add topology features, topology clusters, and topology router.
4. Add Fluency-Correctness Asymmetry to explain weak output routers.
5. Add sensitivity and compute tables.
6. Add 14B and math results.
7. Expand Related Work with 2025 search/overthinking/routing works.
8. Clean limitations and appendices.

Bottom line: The revised submission should look like a substantially strengthened diagnostic paper, not just the original A* paper with extra tables.